MARKET EQUILIBRIUM MODELS

Variables: **Constant (inelastic) demand**

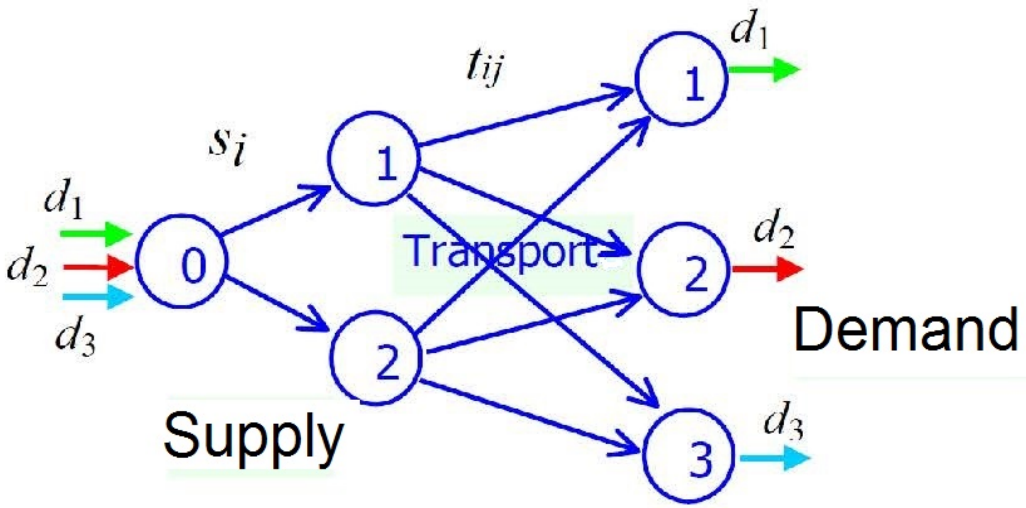
s_i: Amount produced at centre i

t_{ij}: Amount transported on link $i \to j$.

$c_{ij}(t_{ij})$ Unit transportation cost at link $i \to j$ for t_{ij} units

$\pi_i(s_i)$ Unit production cost when s_i units are produced

Constraints:



$$
\begin{aligned}
t_{11} + t_{21} &= d_1 \\
t_{12} + t_{22} &= d_2 \\
t_{13} + t_{23} &= d_3 \\
s_1 + s_2 &= d_1 + d_2 + d_3 \\
s_1 - t_{11} - t_{12} - t_{13} &= 0 \\
s_2 - t_{21} - t_{22} - t_{23} &= 0 \\
s_i \geq 0,\ t_{ij} &\geq 0
\end{aligned}
$$

MARKET EQUILIBRIUM MODELS

t1j >0, t2j>0

$$\pi_1(s_1) + c_{1j}(t_{1j}) = \pi_2(s_2) + c_{2j}(t_{2j})$$

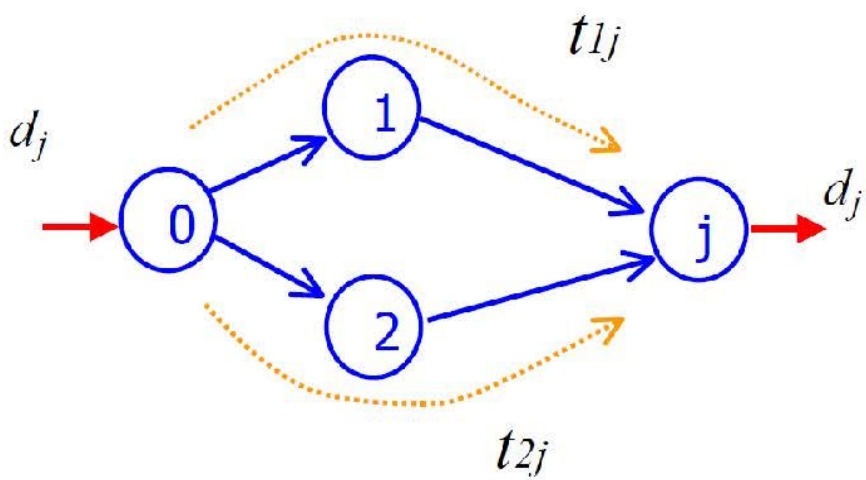
$$\text{o}$$

$$\pi_1(s_1) + c_{1j}(t_{1j}) > \pi_2(s_2) + c_{2j}(t_{2j})$$

$$\Downarrow$$

t2j>0 $\quad t_{1j} = 0$

Constant (inelastic) demand



$$Min_t\, G(t) = F(s(t),t) = \Pi_1(t_{11}+t_{12}+t_{13}) + \Pi_2(t_{21}+t_{22}+t_{23}) + \Sigma_{i,j}\, C_{ij}(t_{ij})$$

$$\begin{array}{ll|l} t_{11}+t_{21} & = d_1 & \lambda_1 \\ t_{12}+t_{22} & = d_2 & \lambda_2 \\ t_{13}+t_{23} & = d_3 & \lambda_3 \\ t_{ij} \geq 0 & & \mu_{ij} \end{array}$$

Karush-Kuhn-Tucker conditions:

$$\frac{\partial G}{\partial t_{ij}} = \pi_i(s_i) + c_{ij}(t_{ij}) = \lambda_j + \mu_{ij}, \quad \mu_{ij} \geq 0, \; t_{ij} \geq 0, \quad \mu_{ij} \cdot t_{ij} = 0$$

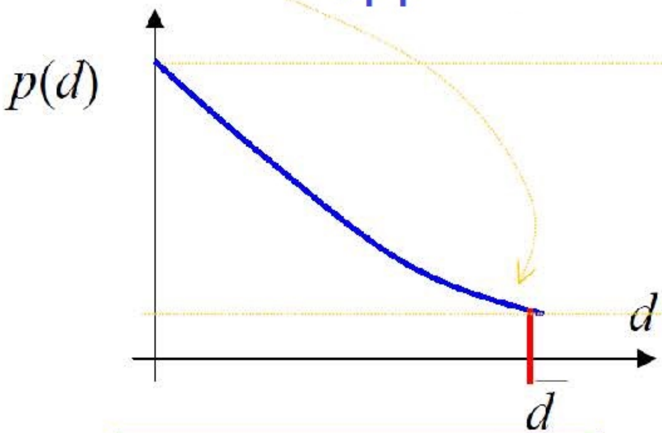
$u_{0j} = Min_{\, i=1,2} \; \{ \; \pi_i(s_i) + c_{ij}(t_{ij}) \; \} = \lambda_j =$ Unit price at market *j*

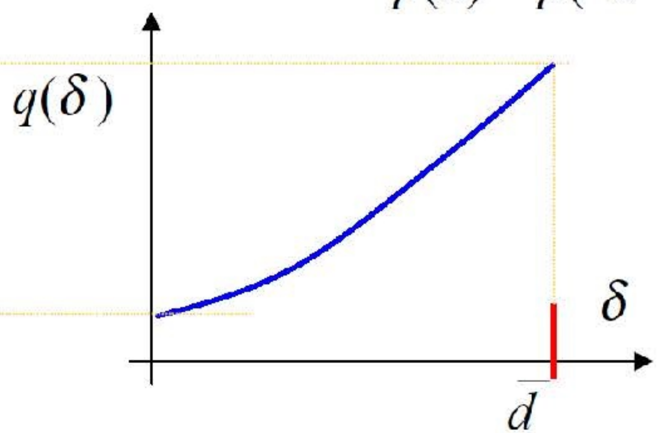
$$[\pi_i(s_i) + c_{ij}(t_{ij}) - u_{0j}] \cdot t_{ij} = 0$$

$$\boxed{\begin{array}{ccc} t_{ij} > 0 & \Rightarrow & \pi_i(s_i) + c_{ij}(t_{ij}) - u_{\mathbf{0}j} = 0 \\ \pi_i(s_i) + c_{ij}(t_{ij}) - u_{\mathbf{0}j} > 0 & \Rightarrow & t_{ij} = 0 \end{array}}$$

$\bar{d}$ = Upper bound on demand

$$p(d) = p(\bar{d} - \delta) = q(\delta)$$



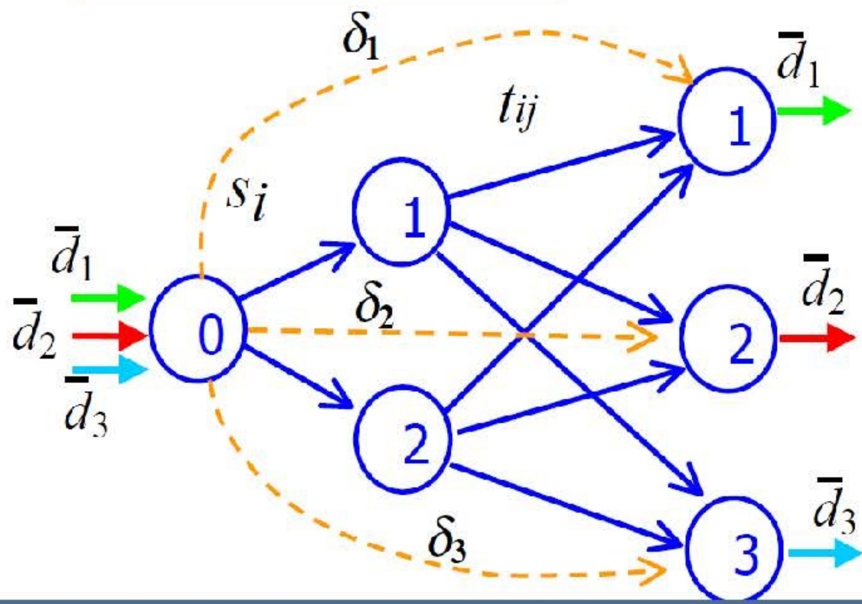


Price $= \mathrm{p}(\text{demand})$

$$\bar{d} = d + \delta$$

Excess demand

Demand consumed by the market



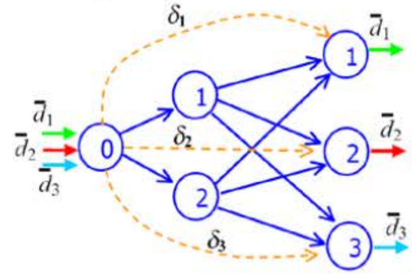
Elastic demand model

$$\Pi_i(s_i) = \int_0^{s_i} \pi_i(x)dx, \quad \Pi_i'(s_i) = \pi_i(s_i), \; i = 1, 2$$

$$Q_j(\delta_j) = \int_0^{\delta_j} q_j(x)dx, \quad Q_j'(\delta_j) = q_j(\delta_j), \; j = 1, 2, 3$$

$$C_{ij}(t_{ij}) = \int_0^{s_i} c_{ij}(x)dx, \quad C_{ij}'(t_{ij}) = c_{ij}(t_{ij}), \quad i = 1, 2, \; j = 1, 2, 3$$

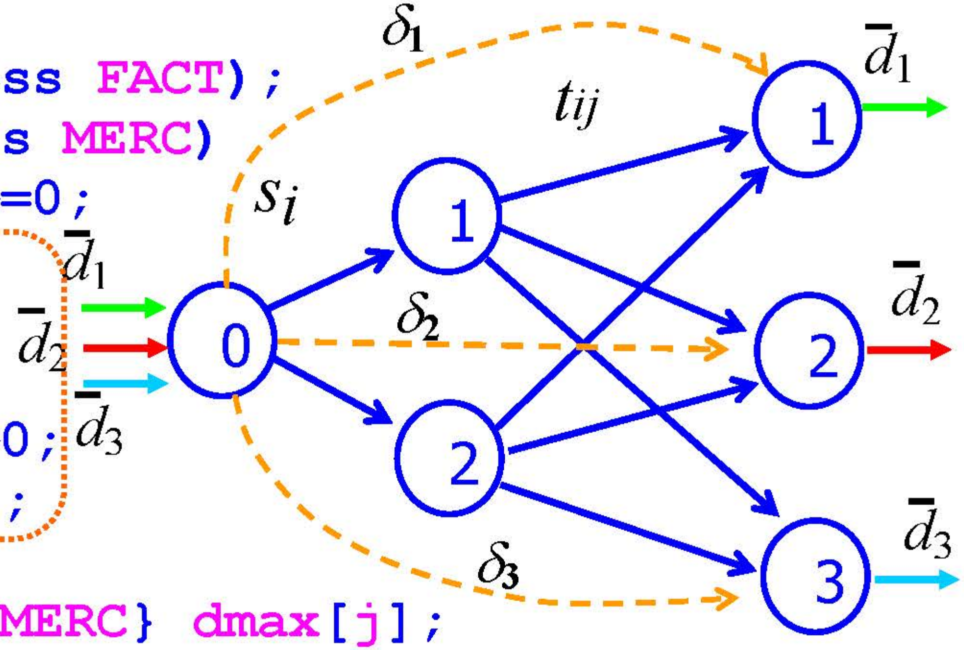
$$Min_{\,s,t}\; F(s,t) = \sum_i \Pi_i(s_i) + \sum_{i,j} C_{ij}(t_{ij}) + \sum_j Q_j(\delta_j)$$



$$t_{11} + t_{21} + \delta_1 = \bar{d}_1$$
$$t_{12} + t_{22} + \delta_2 = \bar{d}_2$$
$$t_{13} + t_{23} + \delta_3 = \bar{d}_3$$
$$s_1 + s_2 + \delta_1 + \delta_2 + \delta_3 = \bar{d}_1 + \bar{d}_2 + \bar{d}_3$$
$$s_1 - t_{11} - t_{12} - t_{13} = 0$$
$$s_2 - t_{21} - t_{22} - t_{23} = 0$$
$$s_i \geq 0, \; \delta_j \geq 0, \; t_{ij} \geq 0$$

```
set FACT;
set MERC;
set ARCTR within (FACT cross MERC);
set ORIGEN;
set ARC_FACT within (ORIGEN cross FACT);
set ARC_EXC within (ORIGEN cross MERC)
param CTRANS {(i,j) in ARCTR} >=0;
param a {(i,j) in ARC_EXC}>=0;
param b {(i,j) in ARC_EXC};
param dmax {j in MERC}>0;
param alfa {(i,j) in ARC_FACT}>0;
param beta {(i,j) in ARC_FACT };

param dtotal default sum {j in MERC} dmax[j];
node O_R {l in ORIGEN} net_out = dtotal;
node P {i in FACT};
node MR {j in MERC} net_in = dmax[j];
arc fict {(l,j) in ARC_EXC} >= 0,
     from O_R[l], to MR[j];
arc xij {(i,j) in ARCTR} >= 0,
     from P[i], to MR[j];
arc si {(i,j) in ARC_FACT} >=0,
     from O_R[i], to P[j];
```



```
minimize F:
sum{(i,j) in ARC_FACT}
   alfa[i,j]*si[i,j]+0,5*beta[i,j]*si[i,j]^2+
sum{(p,q) in ARCTR} CTRANS[p,q]*xij[p,q]+
sum{(r,s) in ARC_EXC}a[r,s]*fict[r,s]+0,5*b[r,s]*fict[r,s]^2;
```